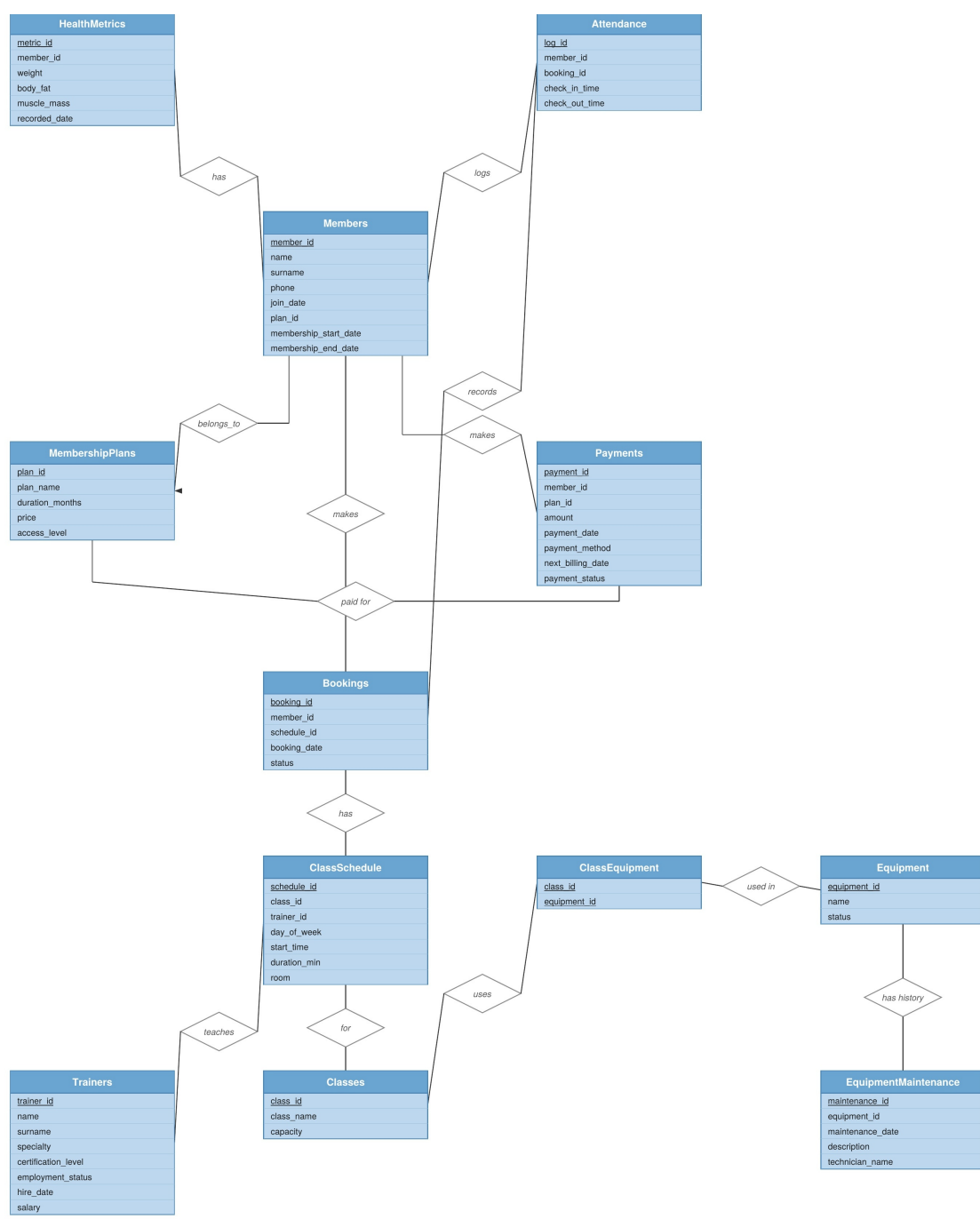


Gym Management System — ER Diagram



Relationships (with Cardinalities)

- belongs_to → members (N) — (1) membership_plans
- has_metrics → members (1) — (N) health_metrics
- makes_payment → members (1) — (N) payments
- makes_booking → members (1) — (N) bookings
- has_booking → bookings (N) — (1) class_schedule
- for_schedule → class_schedule (N) — (1) classes
- teaches → class_schedule (N) — (1) trainers
- logs_attendance → members (1) — (N) attendance
- recorded_by → attendance (N) — (1) bookings
- uses → classes (M) — (N) equipment [via class_equipment]
- has_history → equipment (1) — (N) equipment_maintenance
- paid_for → payments (N) — (1) membership_plans

Transformation Notes

- The belongs_to relationship is a N:1 relationship, therefore plan_id is added as a foreign key in the members table. membership_start_date and membership_end_date are also stored in members.
- The teaches relationship is resolved through class_schedule. trainer_id is added as a foreign key in class_schedule, allowing different trainers for the same class on different days.
- The has_metrics relationship is a 1:N relationship, therefore member_id is added as a foreign key in the health_metrics table.
- The makes_booking and has_booking relationships link members to specific scheduled sessions. member_id and schedule_id are added as foreign keys in the bookings table. Booking is tied to a schedule (not just a class) to enable session-level capacity tracking.
- The logs_attendance relationship is a 1:N relationship. member_id and booking_id are added as foreign keys in the attendance table, linking each check-in to the corresponding booking.
- The makes_payment relationship is a 1:N relationship. member_id and plan_id are added as foreign keys in the payments table. payment_status tracks whether a payment is pending, completed, or failed.
- The uses relationship between Classes and Equipment is a many-to-many (M:N) relationship, resolved using the class_equipment join table with a composite primary key (class_id, equipment_id).
- The has_history relationship is a 1:N relationship. equipment_id is added as a foreign key in the equipment_maintenance table, replacing the single last_maintenance field with a full maintenance history.
- ClassSchedule is introduced to separate class definitions from their scheduled occurrences, storing day_of_week, start_time, duration_min, and room per session.

Final Relation Schema

`membership_plans(plan_id, plan_name, duration_months, price, access_level)`

`trainers(trainer_id, name, surname, specialty, certification_level, employment_status, hire_date, salary)`

`classes(class_id, class_name, capacity)`

`class_schedule(schedule_id, class_id, trainer_id, day_of_week, start_time, duration_min, room)`

`class_id → classes(class_id)`

`trainer_id → trainers(trainer_id)`

`equipment(equipment_id, name, status)`

`equipment_maintenance(maintenance_id, equipment_id, maintenance_date, description, technician_name)`

`equipment_id → equipment(equipment_id)`

`class_equipment(class_id, equipment_id)`

`class_id → classes(class_id)`

`equipment_id → equipment(equipment_id)`

`members(member_id, name, surname, phone, join_date, plan_id, membership_start_date, membership_end_date)`

`plan_id → membership_plans(plan_id)`

`health_metrics(metric_id, member_id, weight, body_fat, muscle_mass, recorded_date)`

`member_id → members(member_id)`

`payments(payment_id, member_id, plan_id, amount, payment_date, payment_method, next_billing_date, payment_status)`

`member_id → members(member_id)`

`plan_id → membership_plans(plan_id)`

`bookings(booking_id, member_id, schedule_id, booking_date, status)`

`member_id → members(member_id)`

`schedule_id → class_schedule(schedule_id)`

`attendance(log_id, member_id, booking_id, check_in_time, check_out_time)`

`member_id → members(member_id)`

`booking_id → bookings(booking_id)`

Additional Notes

- Attendance records represent gym entry and exit of members. Each log is linked to a booking to verify class attendance.
- ClassSchedule separates recurring session details from the class definition, allowing the same class to run on multiple days with different trainers and rooms.
- ClassEquipment is a join table that resolves the M:N relationship between classes and equipment.
- EquipmentMaintenance provides a full audit trail per equipment item, replacing a single last_maintenance date field.
- Payments stores plan_id so each transaction is traceable to the purchased plan.
- Bookings reference schedule_id for session-level precision rather than linking to a class directly.

Irmak Uygun 24070006051

Memnune Arzu Bakır 24070006100

Beyza Mayacı 23070006013

Ulaş Sarıalioğlu 23070006048

Coşkun Doruk Oral 23070006054